# Markus Eder

PhD Researcher — thesis submitted; defence pending
Cellulose chemistry | Analytical methods | Scientific data workflows
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## Profile

Doctoral research completed; thesis submitted; defence pending. Research experience across cellulose chemistry, biomass-derived materials and analytical method development. Combine GC-FID/MS, acidic methanolysis and GPC evaluation with Python-based data workflows and clear technical reporting. Background in environmental engineering, renewable raw materials and biopolymer research. Use human-reviewed AI tools to support technical scripting, documentation and reporting, alongside practical laboratory-tool design.

## Professional Experience

**PhD Researcher | BOKU University** **02/2023–present**
Vienna, Austria

- Developed analytical workflows for lignocellulosic biomass, pulps, cellulose-based materials and coffee-derived samples to support material characterization and technical evaluation.
- Refined methanolysis / GC workflows through derivatization and method comparison, retention-time evaluation and interference tracking.
- Built Python-supported tools connecting analytical files, sample coding, method metadata and CSV/Excel exports; generated grouped summaries, recovery tables, QC-style checks, plots and report-ready outputs.
- Supported GPC and carbonyl-content reporting through grouped evaluation and traceable analytical documentation.
- Communicated findings through technical reports, handouts, posters and conference presentations. Supported colleagues with laboratory setup and supervised one HTL student on practical laboratory support work.

**Research Intern | Polymer Materials Engineering (IMP)** **10/2021–04/2022**
Lyon, France

- Worked on biotechnological production, extraction and characterization of curdlans ($\beta$-1,3-glucans).
- Carried out hydrogel characterization and wet spinning of curdlan fibres in an interdisciplinary laboratory setting.

**Research Intern | Malaviya National Institute of Technology** **08/2016–02/2017**
Jaipur, India

- Investigated biogas digestate using vermicomposting as a resource-recycling route.

## Selected Projects

- **Coffee process-chain characterization:** investigated chemical and structural changes from green coffee beans to spent coffee grounds using acidic methanolysis, total hydrolysis, XRD, TGA and NMR; communicated results in publications and presentations.
- **Methanolysis / GC method development:** developed evaluation templates and method-comparison workflows for complex lignocellulosic samples.
- **Python-based analytical reporting:** linked data and metadata to grouped GPC summaries, recovery tables, plots and structured reporting.
- **AI-assisted workflows and lab tools:** used human-reviewed AI support for scripting, documentation and reporting, alongside OpenSCAD-based laboratory-tool design.

## Technical Skills

**Core analytical workflows:** GC-FID/MS, acidic methanolysis, derivatization, chromatographic method comparison, interference tracking, GPC evaluation and reporting.

**Project-based characterization:** XRD, TGA and NMR in biomass/materials research.

**Materials:** cellulose, hemicellulose, lignocellulosic biomass, pulps, fibre-based materials, coffee-derived residues and curdlan biopolymers.

**Programming and reporting:** Python, structured CSV/Excel exports, sample/metadata linkage, grouped summaries, recovery calculations, LaTeX, Markdown and Microsoft Office.

**Practical workflows:** human-reviewed AI-assisted scripting, documentation and reporting; OpenSCAD-based lab-tool design.

**Basic familiarity:** FTIR, SEM, Origin, MATLAB.

## Education

**PhD research in Chemistry of Renewable Resources** **02/2023–present**
BOKU University, Vienna, Austria

Doctoral research completed; thesis submitted; defence pending. Research focused on cellulose characterization, hemicellulose profiling by methanolysis and fibre-relevant analytical development.

**Dipl.-Ing. in Material and Energetic Exploitation of Renewable Raw Materials (NAWARO)** **09/2019–10/2022**
BOKU University, Vienna, Austria

Specialization in the chemical and material utilization of biomass. Coursework in biobased polymers, natural fibres and life-cycle assessment.

Master's thesis: *Biotechnological Production, Extraction, and Characterization of Curdlans.*

**BEng in Environmental Engineering** **10/2014–09/2019**
Ostbayerische Technische Hochschule Amberg-Weiden, Amberg, Germany

Bachelor's thesis: *Auslegung, Bau und Inbetriebnahme einer Laborwirbelschicht*; design-focused work on a laboratory teaching setup for fluidized-bed experiments.

## Selected Publications

Eder, M.; Bacher, M.; Rosenau, T.; Potthast, A. Tracking transformations of polymeric carbohydrates from green coffee beans to spent coffee grounds: Mannan-dominant structures and valorization implications. *Carbohydrate Polymers* **383** (2026), 125285. doi:10.1016/j.carbpol.2026.125285.

Eder, M.; Grasboeck, S.; van Erven, G.; Bacher, M.; Rosenau, T.; Potthast, A. Reported lignin in coffee and spent coffee grounds is an analytical artefact. *Green Chemistry* **28**(28) (2026), 11867–11876. doi:10.1039/d6gc02448c.

## Languages

German – native | English – professional working proficiency | Hindi – beginner

## Volunteering

DPSG volunteer-leadership and board training in planning, moderation, communication, delegation, conflict handling, teamwork, democratic decision-making and responsibility.